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Disconnected and Sovereign

Building a Private Cloud with Azure Local and M365 Local

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Hey!

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Agenda

Seven parts in 50 minutes, including a video of disconnected operations

01 **Why sovereignty, why now**
Drivers and the Microsoft Sovereign Cloud options

02 **Azure Local as the platform**
Stack, scale, security and the 2026 releases

03 **Fully disconnected or controlled hybrid**
Control plane choices, architecture

04 **Workload placement**
Which workload goes where, and why

05 **Identity and lifecycle without the cloud**
AD FS, PKI, DNS and updates without internet

06 **Microsoft 365 Local on Azure Local**
Architecture, licensing and operational realities

07 **Operate and get started**
Day-2 operations, lessons, path to production

? **Questions**
And ask me anything afterwards

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1

Why sovereignty, why now

Sovereignty is about control, and control comes in levels.



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Digital sovereignty is achieved through choice

Microsoft definition: participate in the digital economy securely, independently and with self-determined controls

MICROSOFT SURVEY, 2025

70%+

of on-premises server decision makers name regulation and compliance as the key barrier to cloud migration

Top reasons: privacy regulation, systems that must stay offline, jurisdictional control and audit control.



Data controls

Where data lives and who can access it: residency, encryption, customer-held keys.



Operational controls

Who operates the platform, from where, and under which jurisdiction.



Disconnected by design

Defence, critical infrastructure and remote sites that cannot depend on internet.



Business continuity

A plan B if the public cloud is unreachable for hours, days or longer.

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Microsoft Sovereign Cloud: three options

Pick the level of control per workload, not per company

Sovereign Public Cloud

Azure and Microsoft 365 in Microsoft datacenters, with extra sovereign controls

- EU Data Boundary and in-region AI processing
- Sovereign Landing Zone with policy guardrails
- Data Guardian and external key management (preview)

Sovereign Private Cloud

Azure-consistent cloud on hardware you own, connected or fully disconnected

- Azure Local with disconnected operations
- Microsoft 365 Local
- Foundry Local for AI on local GPUs

THIS SESSION

National Partner Clouds

Microsoft technology operated by nationally owned partners

- Delos Cloud in Germany (BSI requirements)
- Bleu in France (SecNumCloud)
- In-country operations and procurement

Public or private sovereign cloud?

The questions that decide where a workload belongs

Question	Sovereign Public Cloud	Sovereign Private Cloud
Who owns the hardware and the site?	Microsoft	You
Where does the control plane run?	Azure region	Azure, or a local appliance when disconnected
Can it run without internet?	No	Yes, with disconnected operations
Which services are available?	Full Azure and Microsoft 365	A defined subset of Azure services, plus M365 Local
Who patches and operates it?	Microsoft	You or your partner
Scale and elasticity	Hyperscale	Bound by the hardware you buy

Rule of thumb: use public sovereign controls where you can, and a private sovereign cloud where you must.

2

Azure Local as the platform

The foundation of Microsoft Sovereign Private Cloud.



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Sovereign Private Cloud stack

One platform, connected to Azure or to a local control plane

WORKLOADS

Azure Local VMs	AKS enabled by Arc	Microsoft 365 Local	Foundry Local	Azure Virtual Desktop	GitHub Enterprise Server
GA	GA connected	GA	GA	Connected	Private preview

Connected: Azure control plane
Portal, ARM, Arc, Policy, Defender, Monitor, Update Manager.
Tolerates up to 30 days offline.

Disconnected: local control plane appliance
Portal, ARM, RBAC, Key Vault, Policy and Container Registry on your own management cluster.

Azure Local Compute (Hyper-V) · Storage Spaces Direct or SAN · Networking, SDN and NSGs · Security baseline

Premier Solutions from the Azure Local catalog, required for disconnected operations and M365 Local

DELL Technologies Lenovo Hewlett Packard Enterprise

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Choose your scale

From a single machine to thousands of nodes in one sovereign environment



Small form factor

Compact edge hardware



Hyperconverged

1–16 machines. Rack aware: up to 8 across two rooms



Disaggregated

Compute and SAN scale separately, up to 64 machines



Multi-rack

Integrated racks, hundreds of machines

April 2026: Microsoft states Sovereign Private Cloud now scales to thousands of nodes in a single sovereign environment. Disconnected workload clusters support up to 64 nodes (2604).

What changed since Ignite 2025

The releases that matter for a sovereign design

2601

Rack aware clusters

Two rooms, one cluster, local availability zones

2602

Disconnected operations

Local control plane with multiple Azure Local instances

2604

External SAN storage

Fibre Channel, iSCSI followed in 2607

2604

Local identity with Key Vault

Deploy without Active Directory for the infrastructure

2604

GPU for Azure Local VMs

Full GPU (DDA) and GPU partitioning (GPU-P)

2607 Preview

Confidential VMs

AMD SEV-SNP memory encryption, Linux guests

Secure by default

Security that ships with the platform, not added afterwards

300+ security settings

Baseline enabled at deployment. Aligned with CIS and DISA STIG.

Drift control

Settings refreshed and remediated every 90 minutes.

BitLocker

Data at rest encrypted by default on infrastructure and workload volumes.

Application Control

Only trusted code runs on the hosts. Enforcement mode by default.

Secret rotation

Certificates created and rotated automatically before expiry.

Syslog to your SIEM

CEF or RFC3164 forwarding. Works in disconnected sites.

STANDARDS

Common Criteria
Certified, 23H2 and 24H2 (Jan 2026)

FIPS 140
In process with NIST

Guidance for
ISO 27001, PCI DSS, HIPAA and FedRAMP. You own the compliance, the platform helps.

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3

Fully disconnected or controlled hybrid

Choose the control plane, then design everything around it.



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Four connectivity models

Sovereignty is not binary: decide per site how much the control plane may talk to Azure

1 Connected

Azure is the control plane. Full Arc services: Defender, Monitor, Update Manager, Policy.

Controlled hybrid

2 Connected, outage tolerant

Workloads keep running during intermittent outages of up to 30 days. Resyncs afterwards.

Resilience, not sovereignty

3 Disconnected, limited connectivity

Local control plane. The appliance may reach a few Microsoft endpoints for diagnostics and support.

Local control, cloud support path

4 Disconnected, air-gapped

Local control plane. No connection to Azure. Updates and logs move by import and export.

Maximum isolation

More Azure services → **More local control and more operations work**

Fully disconnected or controlled hybrid?

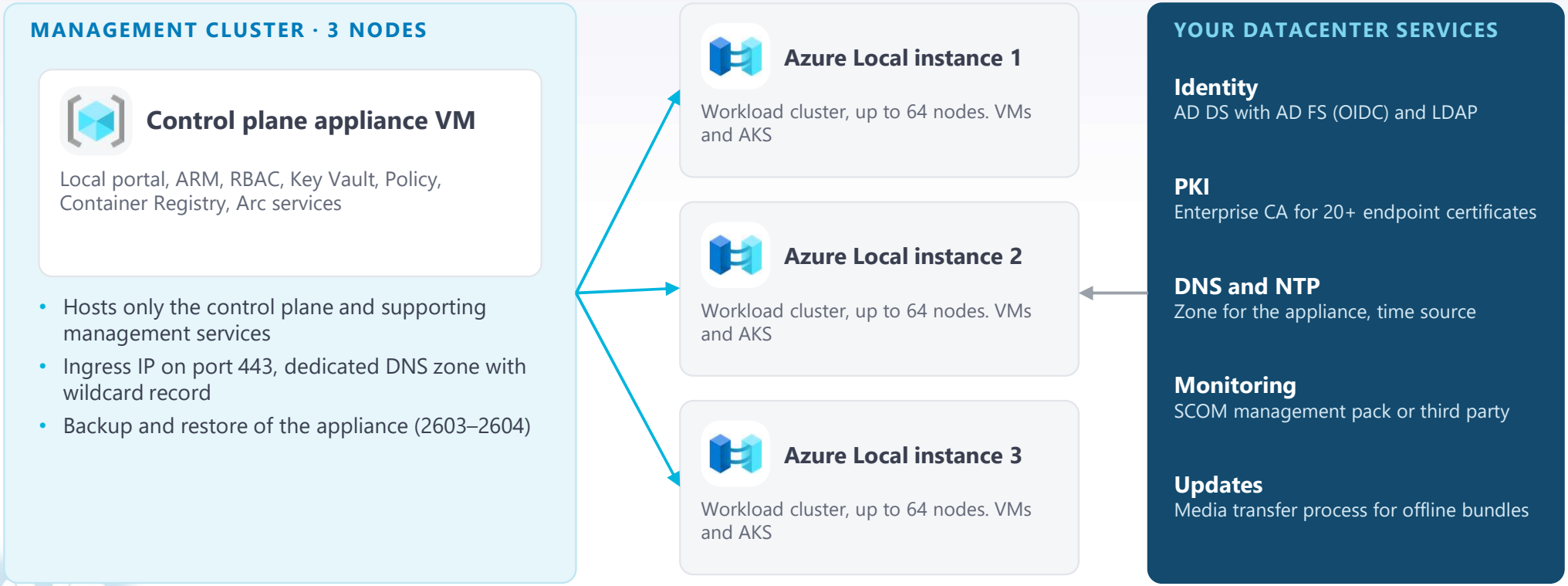
The same Azure Local platform, two very different operating models

	Controlled hybrid (connected)	Fully disconnected
Control plane	Azure: portal, ARM and Azure Arc	Local appliance on a 3-node management cluster
Management identity	Microsoft Entra ID with Azure RBAC	AD DS with AD FS and LDAP, local RBAC
Azure services	Full Arc set: Defender, Monitor, Update Manager, Policy, AVD	Subset only. No AVD, Defender for Cloud, Monitor, Update Manager
Updates	Delivered from Azure and orchestrated for you	Download, transfer, stage and apply release by release
Network	Outbound HTTPS to documented endpoints. 30 days offline tolerance	No connection, or limited for diagnostics and support
Licensing	Host fee per core, or OEM licence	Annual capacity term per core (L3). No AHB or OEM
Hardware	Any Azure Local catalog solution	Premier Solutions, plus the management cluster

Controlled hybrid: workloads and data run locally, Azure holds the control plane and the services you choose. **Disconnected:** no external dependency, and you own every cloud function.

Disconnected operations architecture

A local Azure control plane on its own management cluster



What runs when you are disconnected

A defined subset of Azure, delivered by the local appliance

AVAILABLE ON THE LOCAL CONTROL PLANE

Azure portal	GA	Azure Resource Manager and CLI	GA
RBAC	GA	Managed identity (system-assigned)	GA
Azure Local VMs	GA	Arc-enabled servers	GA
Azure Local device management	GA	Azure Container Registry	GA
Azure Key Vault	GA	Azure Policy	GA
Arc-enabled Kubernetes	Preview	AKS enabled by Arc	Preview

NOT AVAILABLE DISCONNECTED

- Azure Virtual Desktop: brokering runs in Azure
- Microsoft Entra ID: use AD DS with AD FS
- Defender for Cloud and Azure Monitor: use local tools
- Update Manager and Site Recovery
- User-assigned managed identities and management groups

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Sizing the management cluster

Plan the control plane capacity before you order workload hardware

Per node, minimum	Standard	Datacenter
Nodes	3	3
Physical cores	24	24
Memory	128 GB	512 GB
Data drives (SSD or NVMe)	6 × 2 TB	8 × 2 TB
Boot drive	960 GB SSD or NVMe	960 GB SSD or NVMe
Intended scale	100+ managed nodes	1,000+ managed nodes

DESIGN NOTES

- Three-node switchless storage network is recommended
- No tenant workloads on the management cluster
- Proof of concept: four-node configuration, 256 GB per node
- Premier Solutions only: filter the catalog on "Disconnected operations"

Eligibility, licensing and acquisition

Disconnected operations is qualified, not self-service

1 Qualify

Eligible agreement (not MOSA), Standard support or higher, documented business need, skilled team or partner.

2 Pre-qualify

Submit the pre-qualification form through the account team. Microsoft states about 10 business days.

3 Acquire

Create the appliance resource in Azure. Download the manifest and the 250 GB+ appliance package.

4 Deploy

Prepare datacenter services, deploy the appliance, then create Azure Local instances from the local portal.

LICENSING

L3 host fee
Disconnected operations are priced by Microsoft on request (L3).

Two options
Two-month free trial, or a capacity model with an annual term per physical core.

No shortcuts
No Azure Hybrid Benefit and no OEM licence for disconnected deployments.

4

Workload placement

Place each workload by its requirement, not by habit.



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Four questions decide where a workload runs

A simple placement method we use with customers

1. **Can it run in a public region with sovereign controls?**
   Data classification, residency, operator access and jurisdiction.
2. **Must the control plane stay local, or run offline?**
   Law, contract, mission or physical isolation.
3. **Does it depend on cloud-only services?**
   Entra ID, AVD, Defender for Cloud, Azure PaaS, Teams or Copilot.
4. **Can you operate it offline?**
   Update logistics, monitoring, backup, skills and 24/7 ownership.



Sovereign Public Cloud

Yes to question 1

Collaboration, Teams and Copilot, analytics, web and dev/test



Azure Local, controlled hybrid

No to 1, no to 2, or yes to 3

Data that must stay local but can use Azure management: OT and edge, AVD, regulated data



Azure Local, fully disconnected

Yes to 2, no to 3, yes to 4

Classified and mission systems, critical infrastructure, continuity fallback

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Workload placement at a glance

What is available where, status 5 October 2026

Workload	Sovereign Public Cloud	Azure Local, controlled hybrid	Azure Local, fully disconnected
Windows and Linux VMs	● Azure VMs	● Azure Local VMs	● Azure Local VMs
Kubernetes	● AKS	● AKS enabled by Arc	◐ AKS, preview
Exchange, SharePoint, Skype	● Microsoft 365 cloud	● Microsoft 365 Local	● Microsoft 365 Local
Teams and Microsoft 365 Copilot	● Yes	– Not available	– Not available
Virtual desktops	● AVD, Windows 365	● AVD on Azure Local	– Not available
AI inferencing	● Microsoft Foundry	● Foundry Local	● Foundry Local
Security monitoring	● Defender, Sentinel	● Defender, Sentinel	◐ Syslog to local SIEM
Disaster recovery	● Azure Site Recovery	◐ ASR to Azure, preview	◐ Partner backup and replication

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Identity and lifecycle without the cloud

What Entra ID and Windows Update did for you is now your job.



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Identity without Entra ID

AD DS and AD FS become the identity plane for the local control plane

WHAT THE PLATFORM REQUIRES

- AD FS for sign-in (OIDC) and LDAP for group sync
- Universal groups only, one root sync group
- First sync up to 6 hours, then every 15 minutes
- No forced sync, and removed users are not cleaned up
- Only AD and AD FS are validated identity providers

HOW WE SOLVE IT

- Build an isolated AD forest for the sovereign environment: the "bubble"
- Run AD FS highly available. It is now critical for every portal sign-in
- Use secure LDAP (3269) with a read-only service account
- Create the RBAC group structure before deployment
- Use local identity with Key Vault for the Azure Local infrastructure where it fits

Validate before you deploy

```
$idp = New-ApplianceIdentityConfiguration @identityParams
Test-ApplianceExternalIdentityConfigurationDeep -config $idp
```

Certificates and DNS

The appliance exposes more than twenty endpoints, and each needs a trusted certificate

PKI

- Enterprise CA. No self-signed and no public CA
- Certificates valid for at least two years
- CRL distribution point reachable inside the environment
- Portal, ARM, Key Vault, Container Registry, login and more
- Automate requests with the OperationsModule

DNS AND NETWORK

- Dedicated DNS zone, different from your AD domain
- Wildcard A record to the ingress IP, port 443
- Ingress IP in the cluster subnet, outside the deployment range
- Keep 10.131.19.0/24 free: the built-in container range cannot change

```
New-ApplianceExternalCertificatesFromCA -ExternalFQDN $fqdn -CAConfig $ca -OutputFolder $out
nslookup portal.autonomous.cloud.private    # must return the ingress IP
```

Updates without internet

Offline does not mean frozen. You must keep updating to stay supported

1 Download

Get the update package from the appliance resource in Azure, on a connected system

2 Transfer

Move it into the sovereign environment with your approved media or data diode process

3 Stage

Upload and stage on the seed node. Export BitLocker recovery keys first

4 Update appliance

The control plane appliance is always updated first. Automatic rollback on failure

5 Update clusters

Then Azure Local, one release at a time, for example 2602 → 2603 → 2604

Plan it as a process, not a task: owner, maintenance windows, media handling, validation and rollback. Updates can take several hours and the appliance may restart. **Validate the LDAP configuration before every update.**

Lifecycle: everything in the bubble needs a patch path

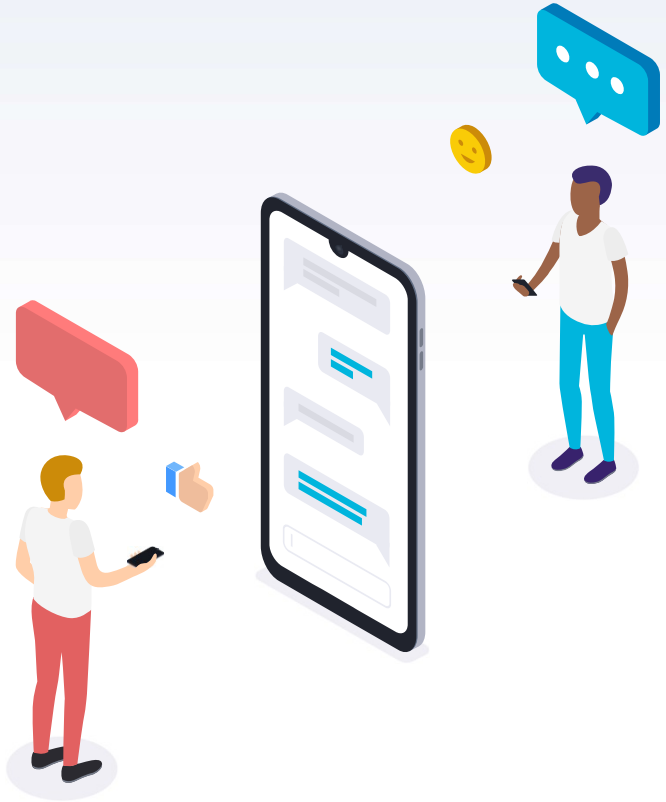
The platform is only one layer. Plan the offline source and owner for each

Layer	Offline source	What to plan
Control plane appliance	Update package from the appliance resource in Azure	Always first. Several hours, may restart
Azure Local instances	Same package, applied release by release	No skipping releases. Cluster-aware, VMs stay up
Firmware and drivers	Premier Solutions hardware vendor	Agree the offline delivery method with the vendor
Windows and Linux guests	WSUS, Configuration Manager or local repositories inside the bubble	Windows Update and public repos are not reachable
Exchange, SharePoint, Skype SE	CUs, SUs and public updates from Microsoft	Modern Lifecycle: stay current to stay supported
Antivirus and signatures	Defender security intelligence from a file share	Daily import is the target, not monthly
Certificates	Your enterprise CA	Track expiry. Appliance certificates need 2+ years validity

6

Microsoft 365 Local on Azure Local

Exchange, SharePoint and Skype for Business inside the sovereign boundary.



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Microsoft 365 Local: what it is and is not

Microsoft server productivity workloads on Azure Local, connected or disconnected

Exchange Server SE

SharePoint Server SE

Skype for Business Server SE

WHAT IT IS

- Subscription Edition servers on Azure Local Premier Solutions
- Validated reference architecture, deployed by a certified partner
- Connected to Azure, or fully disconnected with the local control plane
- Support committed through at least 2035

WHAT IT IS NOT

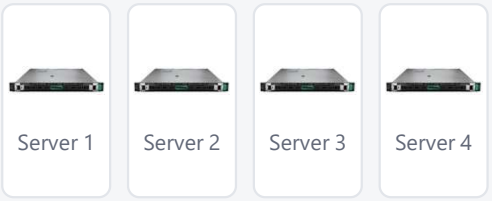
- Not Microsoft Teams, Copilot or other cloud-only features
- Not a replacement for Microsoft 365 in the Sovereign Public Cloud
- Not self-service: planned with a partner and Microsoft
- Entra ID and Intune are not part of the local bundle

Cloud-only services stay in the cloud
Teams and Microsoft 365 Copilot are not part of Microsoft 365 Local.

Reference architecture: large scale

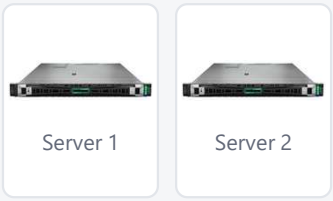
Nine workload servers, plus a management cluster when disconnected

Exchange mailbox



4 × single-node Azure Local

Exchange Edge Transport



2 × single-node Azure Local

SharePoint, Skype and SQL



3-node Azure Local cluster

Disconnected only

3-node management cluster
Control plane appliance instead of Azure. Sized per the management cluster requirements.

Shared services Active Directory and AD FS · Load balancer · Firewall · Separate compute and management networks

Why so many single-node systems? Exchange provides its own availability with database availability groups, so each mailbox server gets its own Azure Local instance. SharePoint and SQL rely on the cluster.

Capabilities compared

What changes compared with cloud and classic on-premises

Capability	Microsoft 365, Sovereign Public Cloud	Microsoft 365 Local	Classic on-premises
MFA and Conditional Access	● Yes	◐ Modern auth through AD FS	◐ Partial
External sharing	● Yes	◐ Needs extra infrastructure	◐ Partial
Meeting recording, Compliance Manager	● Yes	– No	– No
Azure workloads (VMs, Kubernetes, AI)	● Yes	● Yes	– No
Unified control plane	● Yes	● Yes	– No
Full-stack solution validation	● Yes	● Yes	– No
Hybrid and disconnected flexibility	– No	● Yes	◐ Partial

The workloads behave much like classic on-premises servers. The gain is the platform: control plane, validation and Azure services.

Operational realities of Microsoft 365 Local

The workloads are familiar. Running them inside a disconnected boundary is not

Exchange Server SE

- DAG across single-node instances: move databases, then patch one server at a time
- Emergency Mitigation service needs Microsoft endpoints. Disconnected, apply SUs fast instead
- Edge Transport in the perimeter. Mail leaves the boundary only where you allow it

SharePoint Server SE

- Monthly public updates and feature updates twice a year
- Run the configuration wizard on every server after each update
- SQL Server patching and backup are your responsibility

Skype for Business SE

- Cumulative updates applied pool by pool
- No Teams interop or cloud hybrid when disconnected
- Voice, federation and external access need boundary design

COMMON FOR ALL THREE

AD DS and AD FS for modern authentication · Certificates from your enterprise CA · Application-aware partner backup · Monitoring with SCOM or a third-party tool · No Intune for mobile devices

Licensing Microsoft 365 Local

Four licence layers, and they differ between connected and disconnected

Layer	Connected	Disconnected
1. Azure Local host	$10 per physical core per month, or OEM licence	Annual capacity term per core (L3). No OEM licence
2. Windows Server guests	WS subscription $23.3 per core per month, or included in OEM	Windows Server with SA or subscription (Azure Hybrid Benefit)
3. Microsoft 365 servers	Office Server licence + CAL with SA	Office Server licence + CAL with SA
4. SQL Server	Existing SQL Server licensing	Existing SQL Server licensing

Watch out: Azure Hybrid Benefit for the Azure Local host fee does not apply to Microsoft 365 Local. Microsoft 365 E3/E5 eligibility for Microsoft 365 Local was still "to be decided" in June 2026.

7

Operate and get started

Day-2 operations decide if the design survives.



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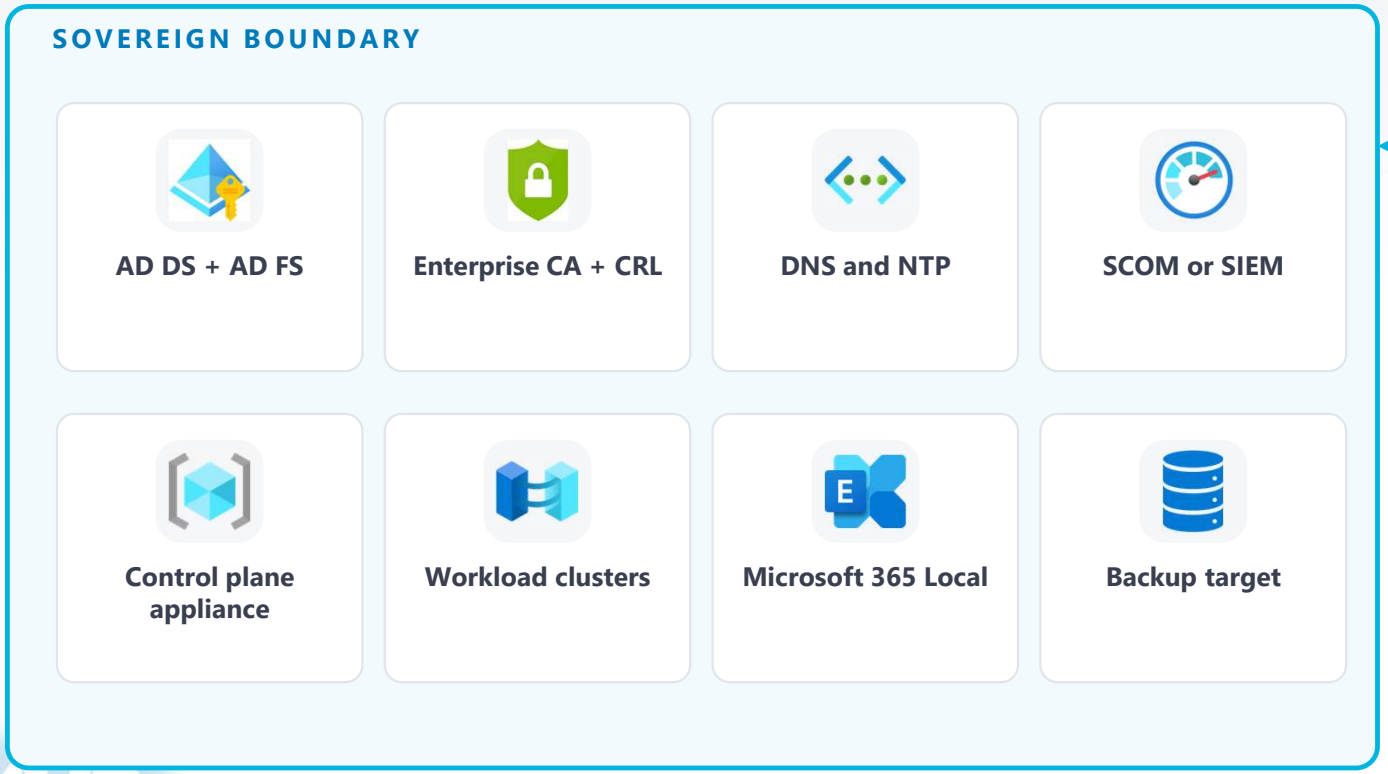
Replace each cloud service you lose

Decide the local replacement before go-live, not after

Cloud capability	Local replacement when disconnected
Azure Monitor and Insights	SCOM 2022/2025 with the disconnected operations management pack, or third-party tools
Microsoft Sentinel	Syslog (CEF or RFC3164) forwarding to your own SIEM
Defender for Cloud	Built-in baseline, drift control, Defender Antivirus, your EDR in guests
Azure Backup and Site Recovery	Partner backup (for example Veeam, Rubrik, Commvault) and appliance backup and restore
Microsoft support upload	On-demand and fallback log collection, exported manually
Kubernetes monitoring	Prometheus and Grafana

Designing the isolated environment

The "bubble": everything the platform needs lives inside the boundary



Controlled crossing

Updates in and logs out only through an approved transfer process or data diode.

Design rules

- No dependency on corporate AD, DNS or PKI
- Separate management and compute networks
- Document every flow that crosses the boundary

An operating rhythm for a disconnected cloud

Recommended cadence. Adapt it to your own risk and compliance requirements

Daily

- Health and certificate alerts
- Backup job status
- SIEM review
- Antivirus signature import

Weekly

- Capacity and performance
- Failed sign-ins and AD FS health
- Prepare next media transfer

Monthly

- Appliance, then Azure Local update
- Guest OS patches
- Exchange, SharePoint, Skype updates

Quarterly

- Restore test: appliance, mailbox, site
- Access review of RBAC groups
- DR exercise

Yearly

- Certificate renewal plan
- Firmware and hardware lifecycle
- Licence term and capacity review

Why the access review: LDAP sync does not clean up removed users. Someone has to own that, every quarter.

Lessons from the field

Six things that decide if a sovereign project succeeds

1 Start with the workload

Check that every workload fits the disconnected service subset before you choose the model.

2 Budget the management cluster

Three extra nodes that run no workloads. Include them in the first quote.

3 PKI and AD FS first

They are on the critical path. Build and test them before the hardware arrives.

4 Make updates a process

Owner, windows, media handling and rollback. Staying current is a support requirement.

5 Accept the feature gaps

No AVD, Entra ID, Defender for Cloud or Azure Monitor when disconnected.

6 Engage early

Pre-qualification, certified partners and licensing take weeks. Start in the planning phase.

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Path to production

A structured approach, aligned with Cloud Adoption Framework thinking

1 Assess

- Classify workloads by sovereignty level
- Public, private or partner cloud per workload
- Confirm the business need

2 Qualify

- Agreement and support plan
- Pre-qualification with Microsoft
- Choose a certified partner

3 Design

- Connectivity model per site
- Identity, PKI, DNS, network
- Sizing incl. management cluster

4 Build

- Premier Solutions hardware
- Datacenter services first
- Appliance, clusters, workloads

5 Operate

- Update and media process
- Monitoring, backup, DR
- Audit trail and handover

Tip: learn the platform on a small connected instance first. The operating model is the same, the control plane is not.

Announced, not shipped

Plan with what is available today, and watch these



Announced

Enhanced change controls

Configurable policies and approval workflows for changes from cloud to edge.



Announced

Site-to-site disaster recovery

Azure Site Recovery between two Azure Local instances.



Announced

Connected to disconnected

Move an existing instance between connected and disconnected operations.



Announced

Sovereign Private Landing Zone

Policy guardrails in Azure to build and test apps for the private subset.



Private preview

GitHub Enterprise Server

Validated GitHub Enterprise Server deployment on Azure Local.



Preview

Confidential computing

Confidential VMs today in preview; confidential GPU announced.

Key takeaways

Four things to remember

01

Sovereignty is an architecture decision

Choose per workload: public sovereign where you can, private where you must.

02

Controlled hybrid is often enough

Fully disconnected only when the requirement is no external dependency at all.

03

M365 Local is server workloads

Exchange, SharePoint and Skype for Business SE on a validated platform, not Teams or Copilot.

04

You own what the cloud did

Identity, PKI, DNS, updates, monitoring and backup. Plan them first, not last.

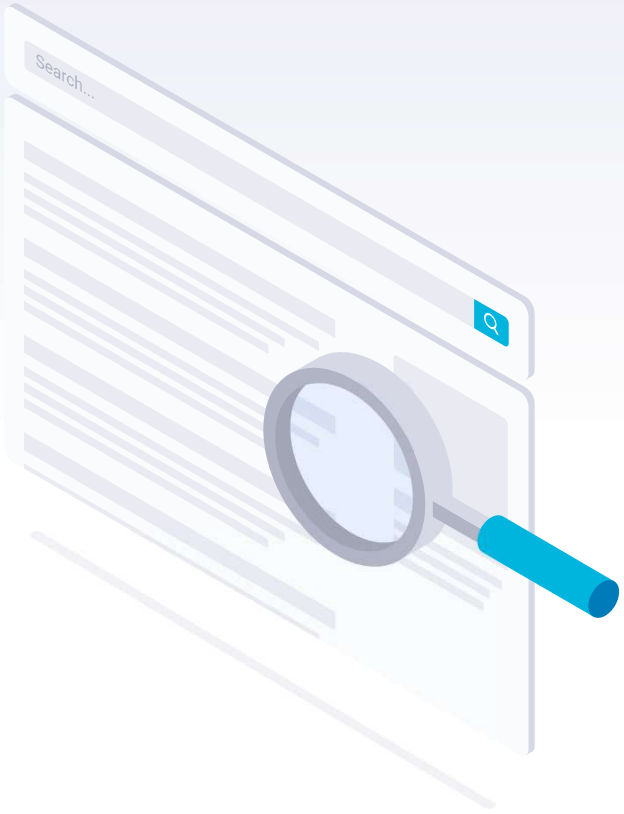
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Thank you!

Questions?

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